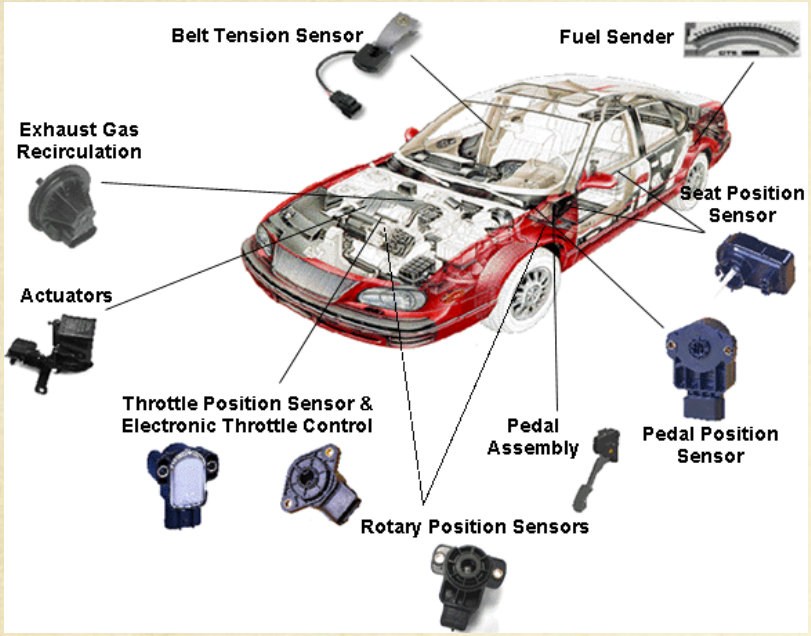
SKEX 2123

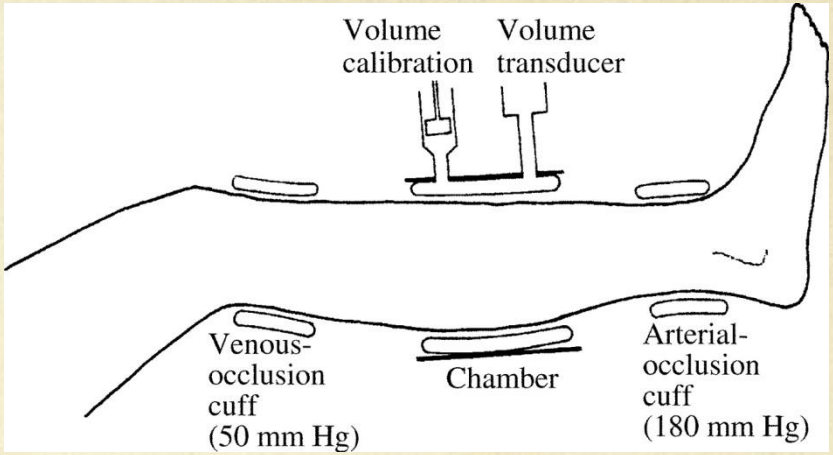
MEASUREMENT AND INSTRUMENTATION

- PROF. IR. DR. HERLINA ABDUL RAHIM

P19a
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Course outline





- This course offers a comprehensive exploration of measurement principles and instruments in electrical engineering. Covering topics from fundamental measurements to advanced sensor technologies and signal processing techniques, the course equips students with the ability to accurately measure and analyze diverse electrical parameters. With a balance of theoretical understanding and hands-on experience through laboratory exercises and projects, students gain proficiency in instrument selection, error analysis, uncertainty estimation, and data processing, enabling them to excel in industries that rely on precise measurements and instrumentation for decision-making and problem-solving.

Course learning outcomes (COs)

CLO1: Explain the principles of basic measurements and instrumentations, concept of metrology and its applications, and basic principle of using transducer in measurement.

CLO2: Analyse the measurements' results based on the electrical measurement.

CLO3: Apply suitable signal conditioning, data acquisition and interfacing approaches for electrical measurement system.

CLO and PLO Mapping

	PLO1	PLO2
CLO1	1,a	
CLO2		1,a,b
CLO3		1,a,b

1 = strong emphasis,
2 = medium emphasis,
3 = low emphasis

a = examinations, tests, quizzes;
b = assignment, report;
c = group presentation, laboratory, seminar;
d = thesis, e = site visit

Student Learning Time

Teaching and Learning Activities	**Student Learning Time (hours)**
1. Face-to-Face Learning	
a. Lecturer-Centered Learning i. Lecture	38
a. Student-Centered Learning (SCL) i. Laboratory/Tutorial ii. Student-centered learning activities	4
1. Self-Directed Learning	
a. Non-face-to-face learning or student-centered learning (SCL) such as manual, assignment, module, e-Learning, etc.	22
a. Revision	38
a. Assessment Preparations	13.5
1. Formal Assessment	
a. Continuous Assessment	2
a. Final Exam	2.5
Total (SLT)	**120**

Week 1	Chapter 1: Introduction to Measurement and Instrumentation (Importance of Electrical Measurements, Metrology)
Week 2	Chapter 1: Introduction to Measurement and Instrumentation (Measurement Units and Standards, Calibration and Traceability)
Week 3	Chapter 2: Measurement Errors and Uncertainty (Types of Measurement Errors, Sources of Errors and Their Analysis)
Week 4	Chapter 2: Measurement Errors and Uncertainty)
Week 5	Chapter 3: Electrical Measuring Instruments (Construction, operation and sensitivity of d'Arsonval meter movement, DC and AC Voltmeter)

Week 6	Chapter 3: Electrical Measuring Instruments (Ammeter, Ohmmeter, Wattmeter)
Week 7	Chapter 3: Electrical Measuring Instruments (Multimeter, Digital Oscilloscope)
Week 8	**Mid-Semester Break**
Week 9	Chapter 4: Basic Transducer (Introduction to Transducer, Temperature Transducer, Resistive Transducer)
Week 10	Chapter 4: Basic Transducer (Pressure, Displacement Transducer)
Week 11	Chapter 5 (Signal Conditioning, Data Acquisition and System Interface): Physical Parameters, Bridges, Signals & Noise and Data in Instrumentation, An Introduction to Signal Conditioning, Components of Signal Conditioning

Week 12	Chapter 5: Data Acquisition Systems (DAQ) Definition and Function, Types of DAQ Systems, Key Components, Data Acquisition, Hardware
Week 13	Chapter 5: System Interface Type of Interfaces. Analog, Digital and Communication Interfaces. Data exchange between devices using various protocols.
Week 14	Project Laboratory Exercises and Hands- on Projects (Electrical measurement, Transducer and data acquisition)
Week 15	

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Course assessment

- Group Assignment 1- 20 % (18 Dec 26)
- Group Assignment 2- 20% (16 Jan 27)
- Individual Assignment - 10% (11 Dec 26)
- Test 1: 20 % (20 Nov 26)
- Test 2: 20 % (09 Jan 27)
- Quizzes x 5- 10%